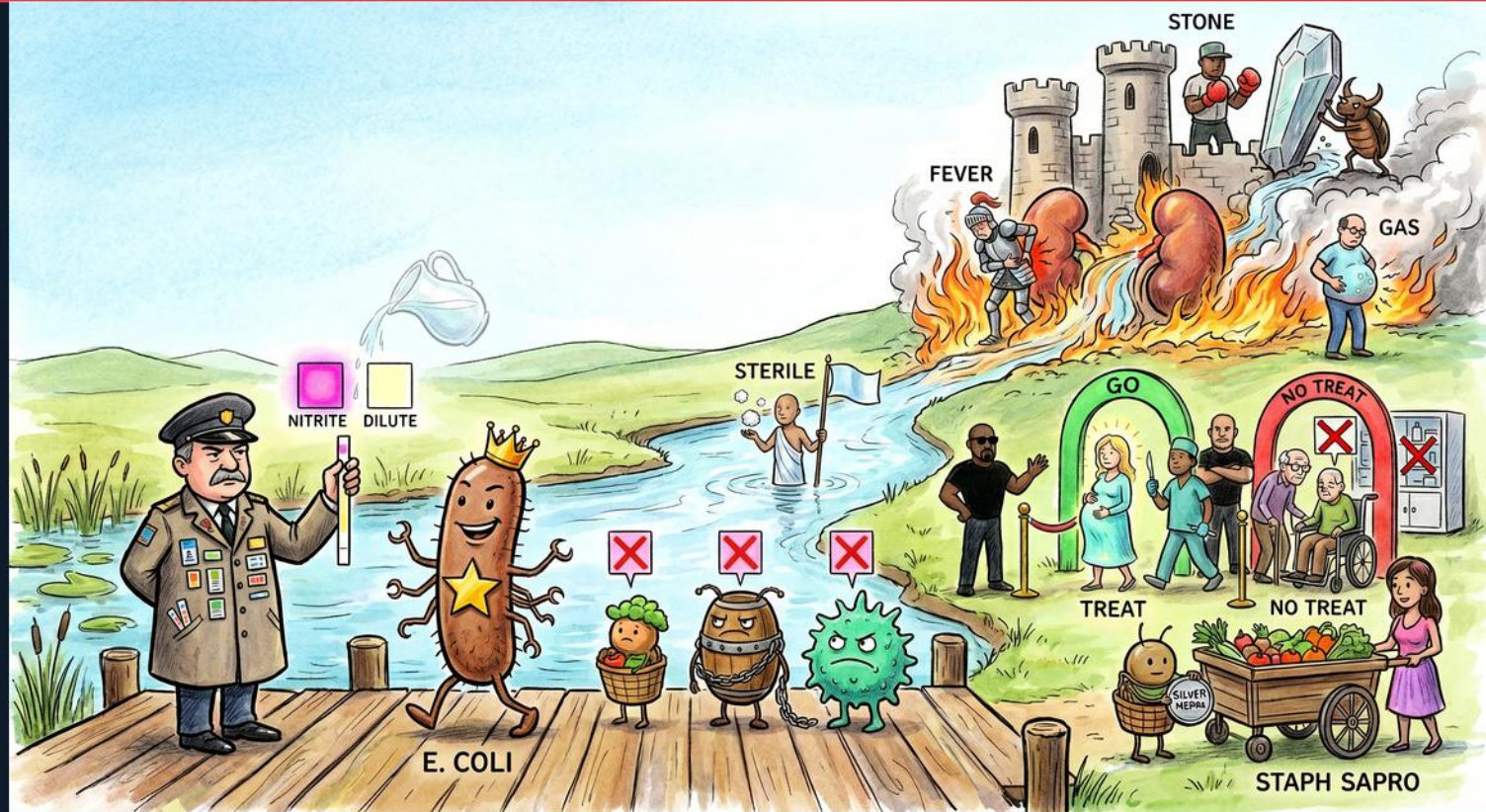


# UTI — Classification & Diagnosis





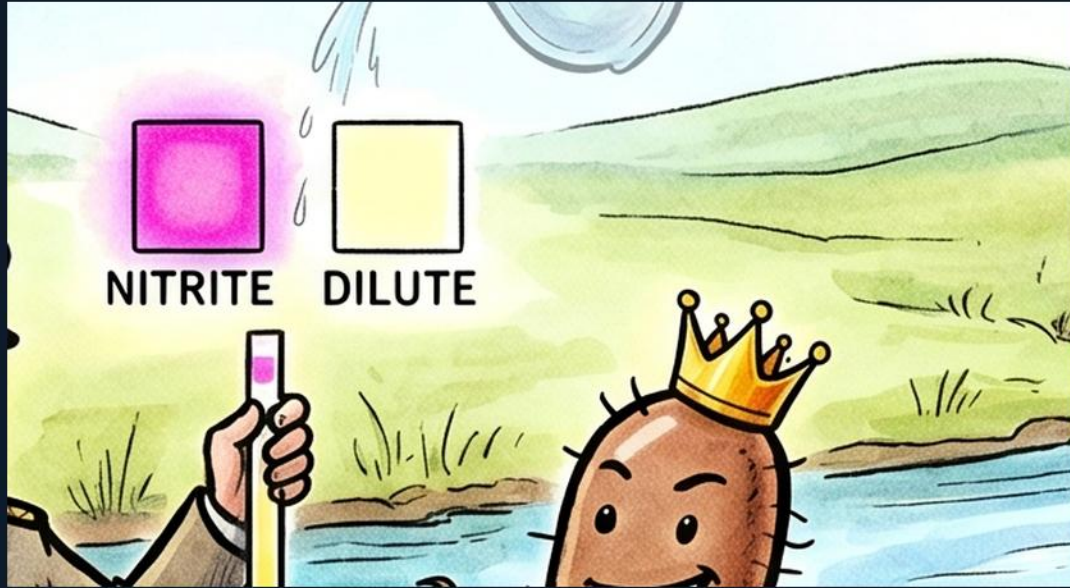
## Nitrite strip (specific)

### WHAT YOU SEE

Inspector holds a bright magenta 'NITRITE' test strip — the glowing positive square is the dipstick turning pink for nitrites

### WHAT IT ENCODES

Positive nitrites are highly SPECIFIC for Enterobacteriaceae (gram-negatives like *E. coli*) that reduce dietary nitrate to nitrite via nitrate reductase. Leukocyte esterase is the SENSITIVE partner marking pyuria (WBCs). A NEGATIVE nitrite does NOT exclude UTI: *Staph saprophyticus*, *Enterococcus*, and *Pseudomonas* lack nitrate reductase. Nitrite also needs ~4 h bladder dwell time, so frequent voiding gives false negatives.



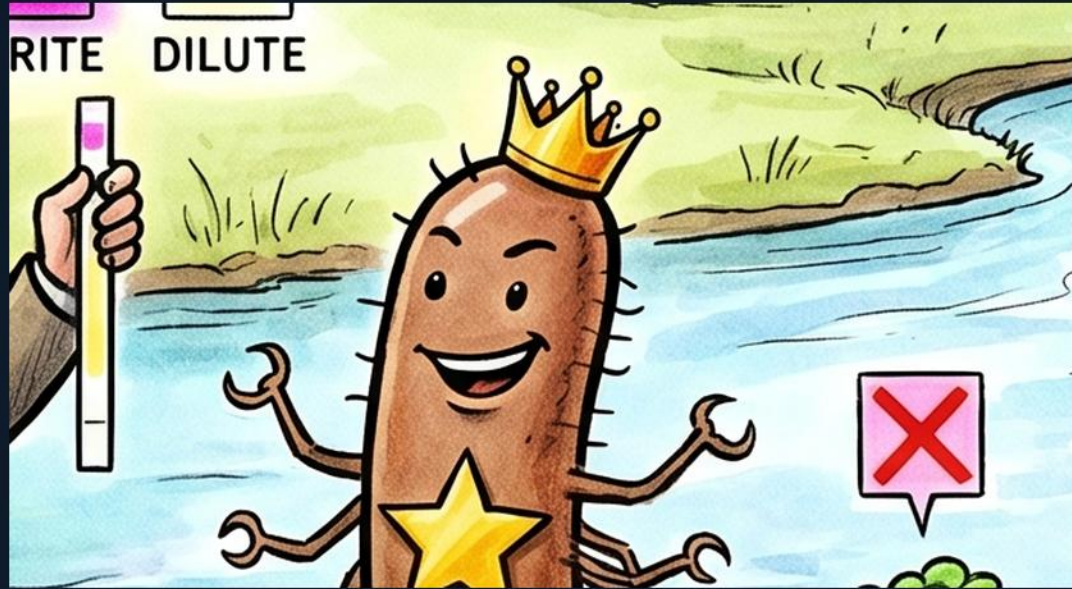
## Dilute = false negative

### WHAT YOU SEE

Faint pale 'DILUTE' strip beside the bright one — washed-out color = a dipstick read on dilute urine

### WHAT IT ENCODES

Very dilute urine (high fluid intake, recent diuretic) drops bacteria, nitrites, and esterase below the dipstick threshold, producing false negatives. A first-morning concentrated specimen is most sensitive. Classic board distractor: a symptomatic patient with a 'negative' dipstick on dilute urine.



## E. coli king (#1 cause)

### WHAT YOU SEE

Crowned bacillus wearing a gold star labeled 'E. COLI' — the king of uropathogens

### WHAT IT ENCODES

Uropathogenic E. coli is the #1 cause of UTI overall — ~80% of uncomplicated cystitis and pyelonephritis. Virulence comes from P-fimbriae/type-1 pili binding uroepithelial receptors to drive ascending infection. Empiric therapy is built around expected E. coli susceptibility, so rising fluoroquinolone and TMP-SMX resistance reshapes first-line choices.



## Red-X contaminants

### WHAT YOU SEE

Three figures each stamped with a red X — vegetable basket, barrel, and spiky virus = organisms NOT typical primary uropathogens / contaminants

### WHAT IT ENCODES

The red-X trio flags non-classic urinary isolates that are usually contaminants or not the primary answer in uncomplicated UTI. Skin/genital flora and viral findings should not anchor the diagnosis when *E. coli* or *Staph saprophyticus* fit the clinical picture better. Always interpret a urine isolate against symptoms and proper collection.

### THE BOARD TRAP

"spiky virus / barrel contaminant" -> wrong; a single coag-negative or environmental isolate in an asymptomatic or improperly collected sample is contamination, not the cause — don't treat it.



## Staph saprophyticus (silver)

### WHAT YOU SEE

Vegetable cart with a SILVER-MEDAL bee labeled 'STAPH SAPRO' — the runner-up cause of cystitis

### WHAT IT ENCODES

*Staphylococcus saprophyticus* is the 2nd most common cause of acute cystitis in young, sexually active women ('honeymoon cystitis'). It is coagulase-NEGATIVE, novobiocin-RESISTANT (vs novobiocin-sensitive *S. epidermidis*), urease-positive, and nitrite-negative, so it can be missed on dipstick.

### THE BOARD TRAP

"contaminant coag-neg Staph" -> wrong; in a young woman with cystitis symptoms *S. saprophyticus* is a true pathogen. Novobiocin RESISTANCE is the buzzword separating it from *S. epidermidis*.



## Sterile flag (sterile pyuria)

### WHAT YOU SEE

Figure in the river holding a plain white 'STERILE' flag — pyuria with a sterile (negative) standard culture

### WHAT IT ENCODES

Pyuria = WBCs in urine ( $\geq 10$  WBC/hpf or positive leukocyte esterase), signaling inflammation but nonspecific. **STERILE** pyuria (pyuria with a **NEGATIVE** standard culture) is a classic board flag pointing **AWAY** from routine UTI: think Chlamydia/gonorrhea urethritis, renal TB, interstitial nephritis, nephrolithiasis, or recently treated UTI.



## TREAT green arch (pregnancy/pre-procedure)

### WHAT YOU SEE

GREEN 'GO/TREAT' archway over a pregnant woman and a procedure team — the only two groups whose asymptomatic bacteriuria you treat

### WHAT IT ENCODES

Asymptomatic bacteriuria is treated **ONLY** in (1) pregnancy and (2) before an invasive urologic procedure with anticipated mucosal trauma/bleeding (e.g., TURP). Mnemonic: 'Pregnant women and pre-procedure — everyone else, leave it alone.' In pregnancy, screen at the first prenatal visit (~12-16 wks) and treat positives 4-7 days with a test-of-cure, since untreated cases progress to pyelonephritis in ~20-30% and raise preterm/low-birth-weight risk.



## NO TREAT red-X arch (elderly/catheter)

### WHAT YOU SEE

RED arch with a big X over elderly and wheelchair figures labeled 'NO TREAT' — asymptomatic bacteriuria you must NOT treat

### WHAT IT ENCODES

Do NOT treat asymptomatic bacteriuria in the elderly, diabetics, spinal-cord-injury, or chronically catheterized patients — even with pyuria. Antibiotics don't help and breed resistance. In a confused elderly patient with a positive urine culture, hunt for another cause before blaming the urine.

### THE BOARD TRAP

"treat the positive urine culture in confused elder" -> wrong; 'treating the culture' instead of finding the real cause is the classic distractor — asymptomatic bacteriuria here is left alone.



## Castle = upper-tract / complicated

### WHAT YOU SEE

Stone CASTLE on the hill above the river — the upper urinary tract / kidneys where infection becomes serious

### WHAT IT ENCODES

Anatomy drives severity: lower tract (cystitis) = local dysuria/frequency/urgency without fever; upper tract (pyelonephritis, the castle) adds fever, rigors, flank/CVA tenderness, and vomiting. Fever and flank pain are the clinical hinge that mandate culture, longer therapy, and tissue-penetrating agents (so NOT nitrofurantoin or fosfomycin).



## FEVER knight + burning kidney

### WHAT YOU SEE

Armored knight labeled 'FEVER' beside a kidney engulfed in flames — febrile pyelonephritis

### WHAT IT ENCODES

Pyelonephritis = systemic infection of the renal parenchyma: fever, rigors, flank/CVA tenderness, nausea/vomiting. Culture is mandatory and therapy is longer with tissue-penetrating drugs. Failure to improve within 48-72 h should trigger imaging to exclude obstruction or abscess.

## STONE



## STONE / obstruction

### WHAT YOU SEE

Boxer punching a giant crystal 'STONE' atop the castle — an obstructing stone behind the infection

### WHAT IT ENCODES

An obstructing stone turns pyelonephritis into an emergency: infected hydronephrosis (pyonephrosis) needs urgent decompression (ureteral stent or percutaneous nephrostomy) plus antibiotics, because antibiotics alone can't sterilize obstructed urine. Obstruction also reclassifies any UTI as complicated.



## GAS man = emphysematous pyelo

### WHAT YOU SEE

Man clutching a gas-swollen flank labeled 'GAS' — emphysematous pyelonephritis

### WHAT IT ENCODES

Emphysematous pyelonephritis is a necrotizing, gas-forming renal infection almost always in DIABETICS (gas from glucose fermentation by *E. coli* or *Klebsiella*). CT showing gas within the parenchyma/collecting system is diagnostic. It is a urologic emergency: IV antibiotics, glycemic control, drainage, and nephrectomy if no response.

### THE BOARD TRAP

"keep giving oral antibiotics" -> wrong; in a diabetic with severe pyelonephritis not improving, get an urgent CT for gas/abscess — high-mortality emphysematous pyelonephritis often needs drainage or nephrectomy.

# Summary — every symbol

## 1. Nitrite strip (specific)

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## 2. Dilute = false negative

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## 3. E. coli king (#1 cause)

Uropathogenic E

## 4. Red-X contaminants

The red-X trio flags non-classic urinary isolates that are usually contaminants or not the primary answer in uncomplicated UTI

## 5. Staph saprophyticus (silver)

Staphylococcus saprophyticus is the 2nd most common cause of acute cystitis in young, sexually active women ('honeymoon cystitis')

## 6. Sterile flag (sterile pyuria)

Pyuria = WBCs in urine ( $\geq 10$  WBC/hpf or positive leukocyte esterase), signaling inflammation but nonspecific

## 7. TREAT green arch (pregnancy/pre-procedure)

Asymptomatic bacteriuria is treated ONLY in (1) pregnancy and (2) before an invasive urologic procedure with anticipated mucosal trauma/bleeding (e.g., TURP)

## 8. NO TREAT red-X arch (elderly/catheter)

Do NOT treat asymptomatic bacteriuria in the elderly, diabetics, spinal-cord-injury, or chronically catheterized patients — even with pyuria

## 9. Castle = upper-tract / complicated

Anatomy drives severity: lower tract (cystitis) = local dysuria/frequency/urgency without fever; upper tract (pyelonephritis, the castle) adds fever, rigors, flank/CVA tenderness, and vomiting

## 10. FEVER knight + burning kidney

Pyelonephritis = systemic infection of the renal parenchyma: fever, rigors, flank/CVA tenderness, nausea/vomiting

## 11. STONE / obstruction

An obstructing stone turns pyelonephritis into an emergency: infected hydronephrosis (pyonephrosis) needs urgent decompression (ureteral stent or percutaneous nephrostomy) plus antibiotics, because antibiotics alone can't sterilize obstructed urine

## 12. GAS man = emphysematous pyelo

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